

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. – 06

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Recap of Previous Lecture



Topic

VLSM





Topics to be Covered



Topic

CIDR

Topic

Routing Table

[MCQ]

#Q. A company has a class C network address of "204 . 204 . 204 . 0". It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

[GATE-IT-2005, 2 Marks]

- ☒ (A) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 192
204 . 204 . 204 . 0 / 255 . 255 . 255 . 128
204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- ☒ (B) 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192
204 . 204 . 204 . 192 / 255 . 255 . 255 . 128
204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- ☒ (C) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
204 . 204 . 204 . 192 / 255 . 255 . 255 . 192
204 . 204 . 204 . 224 / 255 . 255 . 255 . 192
- ☒ (D) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
204 . 204 . 204 . 64 / 255 . 255 . 255 . 192
204 . 204 . 204 . 0 / 255 . 255 . 255 . 192

Ans: D

Class C network (Network Address : 204 . 204 . 204 . 0)

204 . 204 . 204 . 0 0 0 0 0 0 0 0
 8 bit Host ID

Subnet-1 Address : 204 . 204 . 204 . 0 0 0 0 0 0 0 0
 [for 100 host]
 Subnet Mask = 255 . 255 . 255 . 128
 7 bit Host ID

Subnet-2 Address : 204 . 204 . 204 . 1 0 0 0 0 0 0 0
 [for 50 host]
 Subnet Mask = 255 . 255 . 255 . 192
 6 bit Host ID

Subnet-3 Address : 204 . 204 . 204 . 1 1 0 0 0 0 0 0
 [for 50 host]
 Subnet Mask = 255 . 255 . 255 . 192
 6 bit Host ID

Class C network (Network Address : 204 . 204 . 204 . 0)

204 . 204 . 204 . 0 0 0 0 0 0 0 0
 8 bit Host ID

Subnet-1 Address : 204 . 204 . 204 . 1 0 0 0 0 0 0 0
 [for 100 host]
 Subnet Mask = 255 . 255 . 255 . 128
 7 bit Host ID

Subnet-2 Address : 204 . 204 . 204 . 0 0 0 0 0 0 0 0
 [for 50 host]
 Subnet Mask = 255 . 255 . 255 . 192
 6 bit Host ID

Subnet-3 Address : 204 . 204 . 204 . 0 1 0 0 0 0 0 0
 [for 50 host]
 Subnet Mask = 255 . 255 . 255 . 192
 6 bit

Subnet-1 Address	:	204 . 204 . 204 . 128	Subnet Mask	=	255 . 255 . 255 . 128
Subnet-2 Address	:	204 . 204 . 204 . 0	Subnet Mask	=	255 . 255 . 255 . 192
Subnet-3 Address	:	204 . 204 . 204 . 64	Subnet Mask	=	255 . 255 . 255 . 192

Subnet-1 Address	:	204 . 204 . 204 . 128	Subnet Mask	=	255 . 255 . 255 . 128
Subnet-2 Address	:	204 . 204 . 204 . 64	Subnet Mask	=	255 . 255 . 255 . 192
Subnet-3 Address	:	204 . 204 . 204 . 0	Subnet Mask	=	255 . 255 . 255 . 192

Subnet-1 Address	:	204 . 204 . 204 . 0	Subnet Mask	=	255 . 255 . 255 . 128
Subnet-2 Address	:	204 . 204 . 204 . 128	Subnet Mask	=	255 . 255 . 255 . 192
Subnet-3 Address	:	204 . 204 . 204 . 192	Subnet Mask	=	255 . 255 . 255 . 192

Subnet-1 Address	:	204 . 204 . 204 . 0	Subnet Mask	=	255 . 255 . 255 . 128
Subnet-2 Address	:	204 . 204 . 204 . 192	Subnet Mask	=	255 . 255 . 255 . 192
Subnet-3 Address	:	204 . 204 . 204 . 128	Subnet Mask	=	255 . 255 . 255 . 192



Topic : CIDR



- CIDR : Classless Inter-Domain Routing
- IP Address allocation method for IP routing
- Based on Variable-length subnet masking (VLSM)
- Allows flexibility in creating 'supernets'

Example 18 :-

[MSQ]



#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: "155.220.195.0 / 24". An organization request to ISP for range of IP address for its 30 hosts. Which of the following is/are can be a valid (network address) allocation?

- ☒ A 155.220.195.144 / 27
- ☒ B 155.220.195.160 / 27
- ☒ C 155.220.195.192 / 27
- ☒ D 155.220.195.200 / 27

ISP $\rightarrow 2^8$ Address
[8 bit host ID]

2^5 Address
5 bit host ID

Ans: B & C

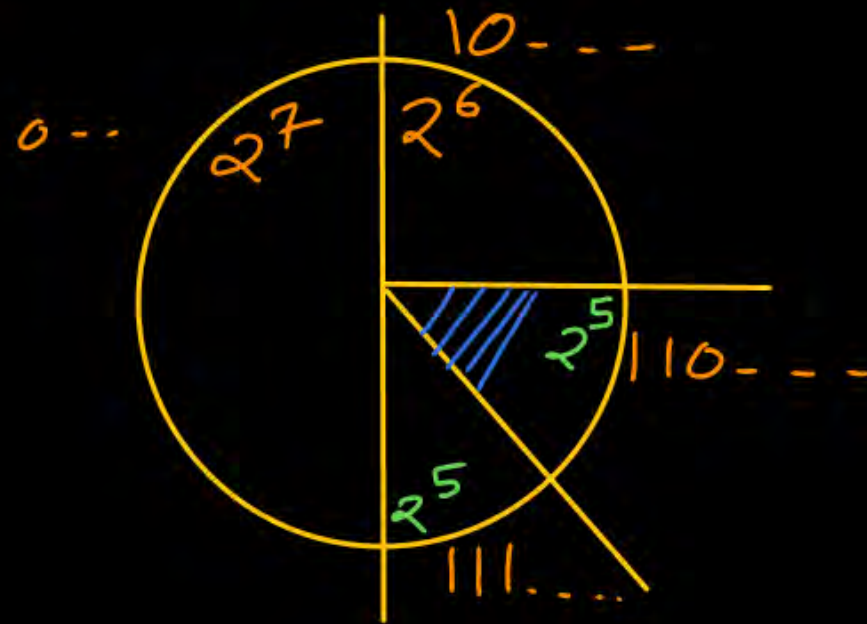
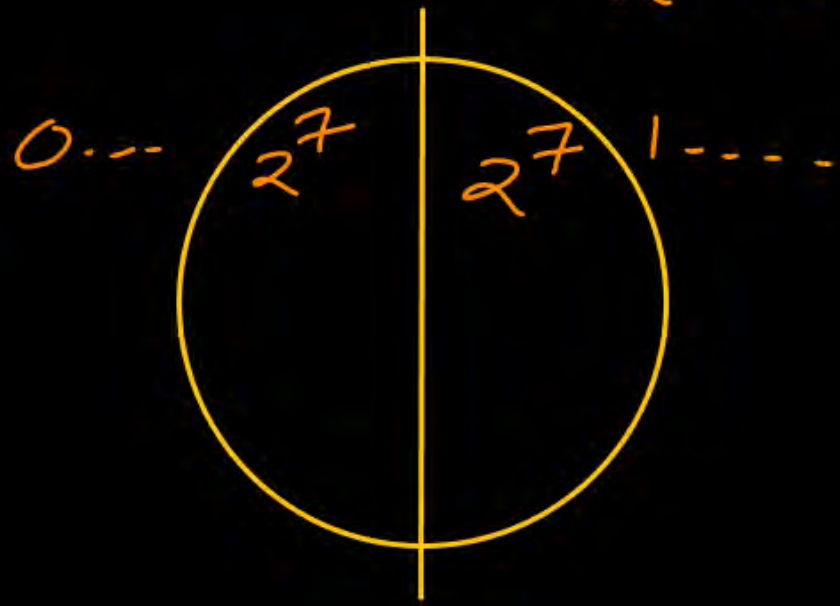
$$144 \rightarrow \overset{b_1 b_2 b_3}{\underline{100}} | \underbrace{0000}$$

$$160 \rightarrow \underline{101} | \underbrace{000000}$$

$$192 \rightarrow \underline{110} | \underbrace{000000}$$

$$200 \rightarrow \underline{1100} | \underbrace{1000}$$

2^8 Address



$$\begin{array}{r} 2^7 \quad 0 \dots \dots \\ \hline 2^6 \quad 10 \dots \dots \\ \hline 2^5 \quad 111 \dots \dots \end{array}$$

Allocate 5-bit host ID

(1) 110...

(2) 111...

($b_1 b_2 b_3$)

(3) 100...

(4) 101...

(5) 000...

(6) 001...

(7) 010...

(8) 011...

Available IP Address space = 155.220.195.0 / 24

155.220.195. ————— / 24
 24 bit prefix 8 bit host ID

Number of hosts
 [for IP address requested] = 30 hosts

Number of bits for host id in allocated chunk = $\lceil \log_2(\text{Number of hosts}) \rceil$ bits
 = $\lceil \log_2(30) \rceil$ bits = 5 bits

Allocated IP Address space = 155.220.195. b₁ b₂ b₃ 0 0 0 0 0 / 27
 27 bit prefix 5 bit host ID

[MCQ] $\Rightarrow$ MSQ

IIT-D, H.W.

[GATE-2020][2 Mark]



#Q. An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space "202 . 61 . 0 . 0 / 17". The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

- I. 202 . 61 . 84 . 0 / 21
- II. 202 . 61 . 104 . 0 / 21
- III. 202 . 61 . 64 . 0 / 21
- IV. 202 . 61 . 144 . 0 / 21

A I and II only

B II and III only

C III and IV only

D I and IV only

[MCQ]

IIT-D, H.W.

[GATE-2012][2 Mark]



#Q. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it : "245 . 248 . 128 . 0 / 20". The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

- A 245 . 248 . 136 . 0 / 21 and 245 . 248 . 128 . 0 / 22
- B 245 . 248 . 128 . 0 / 21 and 245 . 248 . 128 . 0 / 22
- C 245 . 248 . 132 . 0 / 22 and 245 . 248 . 132 . 0 / 21
- D 245 . 248 . 136 . 0 / 24 and 245 . 248 . 132 . 0 / 21

Example 19 :-

[MCQ]



#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses 192.168.0.0 to 192.168.255.255?

✓ A 192.168.0.0 / 16

B 192.168.0.0 / 17

C 192.168.0.0 / 23

D 192.168.0.0 / 24

/16

Ans: A



192.168.0.0 : 192.168.00000000.00000000
192.168.255.255 : 192.168.11111111.11111111

2^{16} Address

CIDR Prefixes : 192.168.00000000.00000000/16
Network Address : 16 bit prefix 16 bit host ID

→ 192.168.0.0/16

[MCQ]

IISC, H.W.

[GATE-2024][2 Mark]



#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- A 10.12.2.0/23
- B 10.12.2.0/24
- C 10.12.0.0/22
- D 10.12.2.0/22

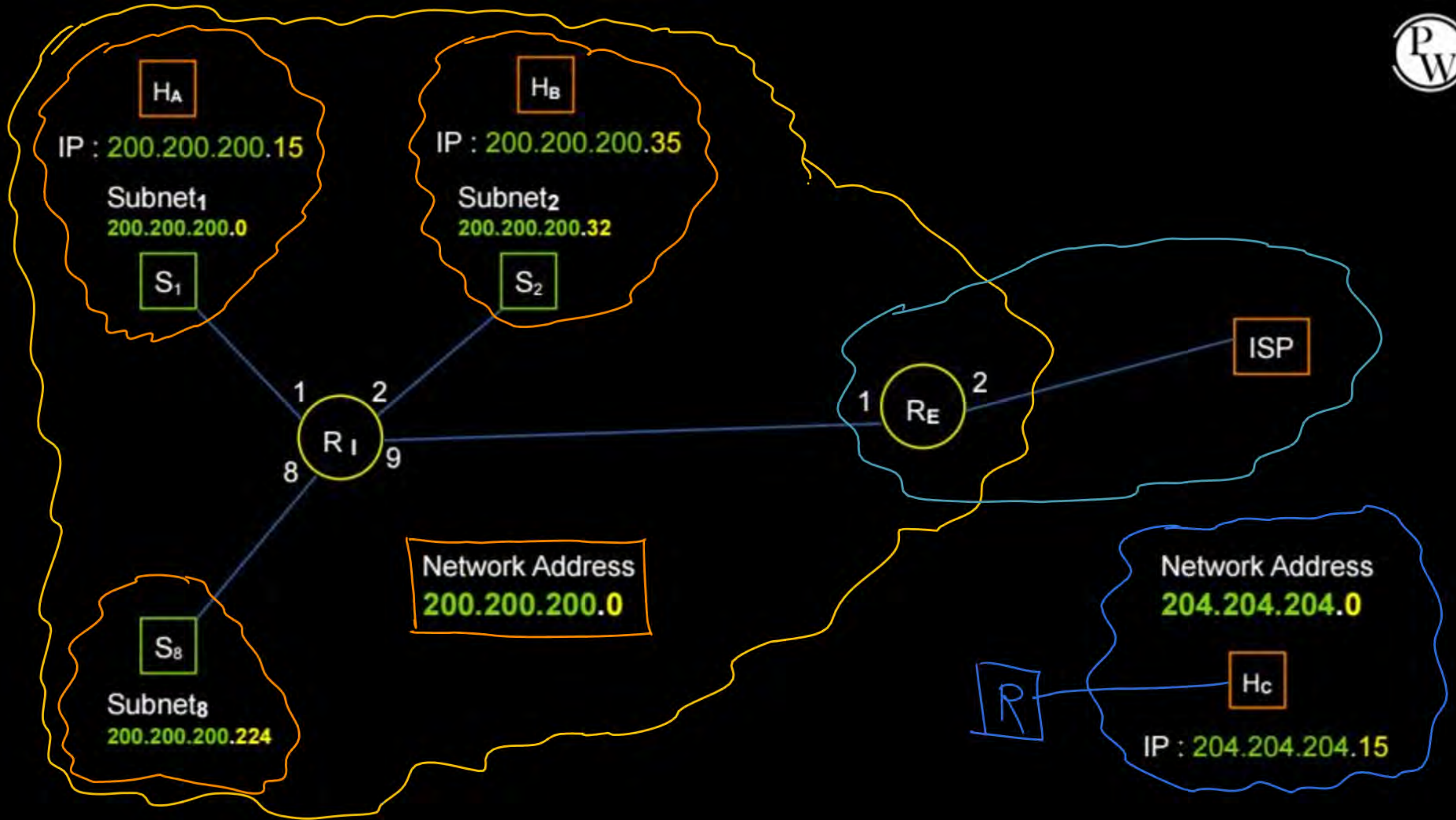
Class C Network :-

Network Address	: 200 . 200 . 200 . 0
Default Netmask	: 255 . 255 . 255 . 0

After 3-bit subnetting :-

Subnet Mask	: 255 . 255 . 255 . 224
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1 st Subnet Address	: 200 . 200 . 200 . 0
2 nd Subnet Address	: 200 . 200 . 200 . 32
3 rd Subnet Address	: 200 . 200 . 200 . 64
4 th Subnet Address	: 200 . 200 . 200 . 96
5 th Subnet Address	: 200 . 200 . 200 . 128
6 th Subnet Address	: 200 . 200 . 200 . 160
7 th Subnet Address	: 200 . 200 . 200 . 192
8 th Subnet Address	: 200 . 200 . 200 . 224





Topic : Forwarding Table



Router (**RE**) forwarding table/ Routing table

Network(Address)	Network(Mask)	Interface ID	Next Hop
200.200.200.0	255.255.255.0	1	R _I
Default		2	ISP



Topic : Forwarding Table



Router (R_1) forwarding table

Subnet(Address)	Subnet(Mask)	Interface ID	Next Hop
200 . 200 . 200 . <u>0</u>	255 . 255 . 255 . <u>224</u>	1	[On Link]
200 . 200 . 200 . <u>32</u>	255 . 255 . 255 . <u>224</u>	2	On Link
200 . 200 . 200 . <u>64</u>	255 . 255 . 255 . <u>224</u>	3	On Link
⋮	⋮		
200 . 200 . 200 . <u>224</u>	255 . 255 . 255 . <u>224</u>	8	On Link
Default		9	[R_E]



2 mins Summary



Topic

CIDR ✓

Topic

Routing Table



THANK - YOU

